

Project Title: Hexa-Ludo: An Innovative Hexagonal Board Game

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Course: Artificial Intelligence
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1. Introduction

The Hexa-Ludo game is a modern digital board game designed as an extension of the classic Ludo, utilizing a hexagonal 24-tile board with added features like power-up tiles and Minimax-based AI. The game supports Player vs Player and Player vs AI modes, offering a strategic and engaging experience.

2. Objective

- To develop a 2D hexagonal board game using Pygame.
- To implement game logic involving turn-based movement, dice rolls, and tile effects.
- To integrate AI using the Minimax algorithm with Alpha-Beta pruning.
- To design an engaging user interface with animations and visual feedback.

3. Technologies Used

- Language: Python 3
- Library: Pygame for game development and GUI
- Algorithms: Minimax with Alpha-Beta Pruning (for AI moves)

4. Game Design

4.1 Board Structure:

- A 5x5 hexagonal grid is created to represent 24 valid tiles.
- Power-Up Tiles: Special tiles (e.g., tile numbers 3, 6, 9...) trigger effects when landed on.

4.2 Players:

- Three players: Player 1 (Green), Player 2 (Blue), and AI (Red).
- Blocking Rule: Players can push others back by one tile.

4.3 Power-Ups:

- Extra dice roll or move forward one tile.

4.4 Dice Roll:

- Animated dice rolls (1 to 6).

5. Artificial Intelligence

5.1 AI Logic:

- AI (Player 3) uses the Minimax algorithm.
- Plans moves to simulate optimal play.
- Currently uses a random heuristic function.

5.2 Simulated Functions:

- `evaluate_board()`, `get_valid_moves()`, `simulate_move()`, `undo_move()`.

6. Results

- Successfully implements a turn-based Hexa-Ludo game.
- Includes AI turns, animations, and tile-based power-ups.
- Visual feedback includes colored tokens, hexagonal board, and win screen animation.

7. Challenges

- Accurate hexagonal positioning.
- Designing a strong AI evaluation function.
- Ensuring smooth transitions between turns.

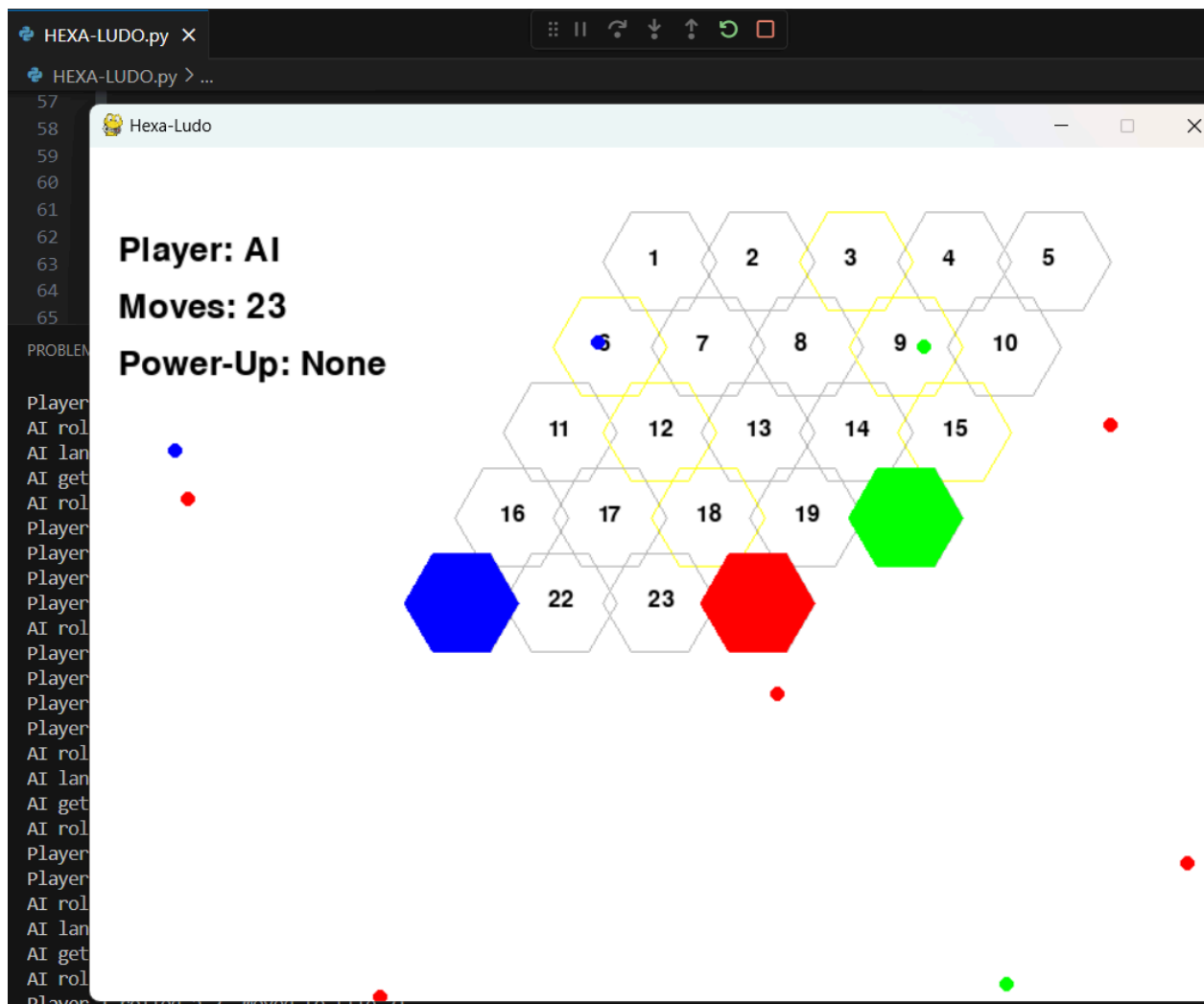
8. Future Work

- Add menu for game mode selection.
- Improve AI evaluation heuristics.

- Enable networked multiplayer.
- Add animations for token movement.

9. Conclusion

Hexa-Ludo is a modern board game with AI and strategic complexity. It offers a fun and engaging user experience and lays the foundation for future improvements and expansion into a full-featured game.



Player 2 gets to move an extra space!
AI rolled a 2, moved to tile 3
AI landed on a power-up tile 3!
AI gets an extra roll!
AI rolled a 2, moved to tile 5
Player 1 rolled a 5, moved to tile 9
Player 1 landed on a power-up tile 9!
Player 1 gets to move an extra space!
Player 2 rolled a 3, moved to tile 7
AI rolled a 6, moved to tile 11
Player 1 rolled a 5, moved to tile 15
Player 1 landed on a power-up tile 15!
Player 1 gets to move an extra space!
Player 2 rolled a 3, moved to tile 10
AI rolled a 1, moved to tile 12
AI landed on a power-up tile 12!
AI gets an extra roll!
AI rolled a 5, moved to tile 17
Player 1 rolled a 3, moved to tile 19
Player 2 rolled a 6, moved to tile 16
AI rolled a 1, moved to tile 18
AI landed on a power-up tile 18!
AI gets an extra roll!
AI rolled a 5, moved to tile 23
Player 1 rolled a 2, moved to tile 21
Player 2 rolled a 5, moved to tile 21
AI rolled a 1, moved to tile 24